Design Brief – Review 2

0. Header

Project code	FA26SE036
Project name	Develop an AI-Assisted Omnichannel Order Fulfillment and Packaging Optimization System (OptiPack AI)
Team	OptiPack AI – Nguyễn Phương Mỹ Thuận (leader), Lê Đức Trung Thi, Huỳnh Quốc Việt, Phan Huỳnh Hải Phượng
Supervisor	Thân Thị Ngọc Vân
Repository link (tag review-2)	https://github.com/ThuaanNguyeen111/OptiPackAI (tag review-2)
Version	1.2
Date	08/11/2026

1. Scope

1.1 Registration Changes

Section	Current	Proposed	Reason
3.1 Order synchronization	Facebook Marketplace, TikTok Shop, Shopee, Lazada, website (optional)	Lazada + AURELLE company website (full sync); TikTok Shop, Shopee, Facebook Marketplace removed	Partner API access constraints (see BM03 #1)
3.2 System Actors – Store Owner	Connect e-commerce platforms	Admin connects shops; Store Owner views connection status	OAuth gives full access to the seller account (BM03 #4)
3.1 Dashboard & Analytics	Dashboards on channels, fulfilment, shipping, cost; Excel / PDF / CSV export	Packing reports, packaging savings, action history with CSV export; Excel / PDF export removed	BM03 #2
3.2 Suggested technologies	Spring Boot, PostgreSQL, Apache Kafka (suggested)	NestJS, MongoDB, scheduled jobs + webhooks + outbox; ReactJS, Redis, Docker, OR-Tools, LLM API (Groq) kept; Next.js added	BM03 #3

1.2 Out of Scope

Item	Reason
Order synchronization for Tiki, TikTok Shop, Shopee, Facebook Marketplace	Partner API access constraints; the TikTok Shop adapter file is not registered. Tiki: the team is registering a tax code for a Tiki seller account; the Tiki adapter added on 31/10/2026 is outside the Review 2 demo
Write-back of order status to Lazada	Only AURELLE receives status and stock write-back
Carriers other than GHN as live integration	GHN (staging environment) is the only carrier with a live API; other carriers use sample rate tables
Excel / PDF analytical report export	Only CSV export of the action history; the home dashboard shows live figures on screen only

2. System at a Glance

Online sellers who sell on several channels handle orders in separate tools, pick from memory, choose boxes by habit and book carriers by hand. OptiPack AI gives one seller and its warehouse team a single system: it imports orders from Lazada and the AURELLE web store, groups orders of the same recipient, reserves stock and guides picking by bin and oldest lot, computes a 3D packing plan per order (box, item placement, fillers), books one GHN shipment per parcel, follows delivery and returns, and writes status back so customers can track their orders. Core value: fewer picking and packing errors, lower packaging and shipping cost, and one traceable order lifecycle. Architecture diagram: SDD (Report 4), section 1.1.

3. Key Design Decisions

ID	Decision	Options considered	Chosen option and rationale	Owner	Related FR / diagram
D1	Stack and database	NestJS + MongoDB; Spring Boot + PostgreSQL	NestJS + MongoDB: flexible order documents, transactions for stock and status changes, one language across back end and front end	Thuận	ERD; SDD 1.3
D2	Packing optimization	Pure heuristic; solver service; hybrid	Hybrid: built-in BRKGA heuristic always available, Python OR-Tools CP-SAT proves plans, fallback if the service is down	Thi	WF-03; SDD 4.5
D3	Order consolidation	Per order; per recipient group	Order group per recipient key (platform, phone, address, city); split and merge before picking starts	Thuận	WF-01
D4	Marketplace integration	Webhook only; polling only; adapter per channel	Adapter per channel; Lazada polled by a scheduler every minute with a per-shop interval (default 5 min), AURELLE by signed webhooks with polling backup	Thuận	WF-01; SDD 4.2
D5	Status write-back and stock integrity	Direct calls; outbox queue	Outbox written in the same transaction, retried with back-off; FIFO stock lots and idempotent scans	Thuận	WF-02, WF-04, WF-05

4. Top Risks

ID	Risk	Impact	How we verify or mitigate	Owner
R1	No staging environment for the demo	Review 2 requires demos on staging; local demo is not accepted	Deploy the Docker Compose / Kubernetes setup to a hosted staging and seed data with the repository scripts	Thuận
R2	Dependence on external APIs (Lazada, GHN)	Sync or booking stops when an API changes or is down	Adapters with mock mode, retries with back-off, idempotent events, contract tests with mock servers	Thuận
R3	3D packing quality and solver availability	Wrong box choice, or no plan when the solver is down	Weight tolerance and packer review, fallback heuristic, benchmark scripts compare solvers	Thi

5. Known Gaps

ID	Gap	Location	Plan	Owner	Due
G1	Mobile app (registration 3.2, FE-06) not implemented; scanning is done on the web portal	Registration 3.2; Report 1 FE-06; SDD 1.1; BM02 F01	Build the Flutter app in Iteration 3 (Report 2, WBS 4.2)	Việt	20/11/2026
G2	Home dashboard widgets (revenue by channel, finance, staff) show sample data; no back-end aggregate API	fe/src/pages/ DashboardPage.tsx; registration 3.1	Done 08/11/2026: the dashboard reads GET /dashboard/summary.	Thuận	20/11/2026
G3	No hosted staging environment for the demo	docs/ DEPLOYMENT.md; SDD 1.3 d	Deploy Docker Compose / Kubernetes to a hosted server and seed demo data with repository scripts	Thuận	Before Defense 1.1
G4	Automated back-end tests use mocked database models; no end-to-end test on a real	be/src/**/*.spec.ts	Add end-to-end tests for pick → pack → ship → return	Thi	04/12/2026

ID	Gap	Location	Plan	Owner	Due
	MongoDB replica set				
G5	Collection packaging_recommendations remains in the schema although the packing flow uses packing_plans	be/src/modules/packaging/schemas; ERD	Keep it only as migration history and mark it legacy in the ERD, or remove it	Thi	20/11/2026
G6	Registration: merge orders across platforms; the system groups orders only within one shop (BM02 F02, Kept)	Registration 3.1; SRS BR-01	Kept: each platform has its own carrier order; same-recipient groups can be linked (SRS UC-22 "Link / Unlink groups manually").	Thuận	—
G7	No state machine for Packing Plan, Shipment, Return Request, Order (BM02 F03)	SDD 3.8	In progress: Order state machine added 08/11/2026 (SDD Figure 12, draw.io); Order Group, Packing Plan, Shipment and Return Request still to draw.	Thi	15/11/2026
G8	Architecture diagram shows a mobile app; misses AURELLE website, Open API simulator, Google OAuth, SMTP (BM02 F04)	SDD Figure 1	In progress: redrawn by hand in draw.io (SDD Figure 1); Google OAuth and SMTP still to add to the figure.	Thi	15/11/2026
G9	Logical ERD misses main entities of the SRS entity table (BM02 F05)	SDD Figure 3	Add all main collections to the ERD; add a database script	Việt	15/11/2026

6. Proposed Mandatory Features

ID	Feature	Actor	Acceptance criteria (testable)	Test	Status
F-01	Authentication and role-based access	Staff (5 roles)	5 failed logins lock the account for 15 min; a role cannot open screens or APIs outside its permissions (403)	TC-F01-01 … TC-F01-03 (Report 5)	Done
F-02	Marketplace connection and order sync	Admin, System Scheduler	A new Lazada order appears in OptiPack within the shop sync interval + 1 min (default ≤ 6 min); a duplicate event creates no second order	TC-F02-01 … TC-F02-03 (Report 5)	Done
F-03	Order consolidation and groups	Store Owner, Shipping Coordinator	Orders with the same recipient key join one open group; a merge after a scan is rejected	TC-F03-01 … TC-F03-03 (Report 5)	Done
F-04	Stock reservation and FIFO lots	Warehouse Staff	Picking consumes the oldest lot first; bin stock never becomes negative	TC-F04-01 … TC-F04-03 (Report 5)	Done
F-05	Picking with missing-item handling	Warehouse Staff, Packaging Staff	A scan over the ordered quantity is rejected; a reported missing item stops the group until a decision	TC-F05-01 … TC-F05-03 (Report 5)	Done
F-06	3D packing plan, approval and packing	Packaging Staff	Every item is placed or the plan cannot be approved; a parcel outside weight tolerance is held for review by another person	TC-F06-01 … TC-F06-03 (Report 5)	Done
F-07	Shipping with GHN	Shipping Coordinator	One GHN order per parcel; a GHN return (or, without carrier API, a second failed attempt) moves the shipment to return to the warehouse	TC-F07-01 … TC-F07-03 (Report 5)	Done

ID	Feature	Actor	Acceptance criteria (testable)	Test	Status
F-08	Returns and exchanges	Customer, Store Owner, Warehouse Staff	A request after 15 days is refused; the creator of a ticket cannot approve it; an exchange creates a replacement order	TC-F08-01 … TC-F08-03 (Report 5)	Done
F-09	AURELLE web store and order tracking	Customer	A placed order shows a 7-step progress bar that follows warehouse and carrier events within 1 minute	TC-F09-01 … TC-F09-03 (Report 5)	Done
F-10	Documents and labels	Warehouse, Packaging Staff, Shipping Coordinator	Packing slip, shipping label, picking list, manifest and bin labels download as PDF	TC-F10-01 … TC-F10-03 (Report 5)	Done

7. Non-Functional Requirements

ID	Quality attribute	Metric	Threshold	Measurement method
NFR-1	Performance	Time from Lazada order creation to appearing in OptiPack	≤ sync interval + 1 min (default ≤ 6 min)	Create a test order and compare with the sync job log (TC-NFR1, manual)
NFR-2	Performance	Packing plan computation time per order	BRKGA p95 ≤ 1 s; CP-SAT proof ≤ 6 s	be/scripts/solver-benchmark.ts and cpsat-benchmark.ts on 300 orders (TC-NFR2)
NFR-3	Security	Failed logins before lockout; role checks	Lock after 5 failures for 15 min; 100% of staff endpoints return 403 to an unauthorized role	Automated auth / role-guard tests (TC-NFR3, TC-F01-02, TC-F01-03)
NFR-4	Reliability	Status write-back success	Up to 8 retries with back-off; 0 lost changes	Outbox test: AURELLE returns errors, job retries until the limit (TC-NFR4)
NFR-5	Data integrity	Stock correctness under repeated / concurrent scans	0 negative bin stock; a repeated scan with the same client event id is not counted twice	Send the same pick scan twice with the same client_event_id (TC-NFR5)
NFR-6	Availability	Packing without the optimization service	Plan still produced when the CP-SAT service is down (label heuristic)	Stop the packer service and compute a plan (TC-NFR6)

8. Ownership Map

Member	Role	Owned parts (2–3)	Main workflow owned	Key decisions / PRs
Nguyễn Phương Mỹ Thuận	Leader	Marketplace integration and order intake; GHN shipping and AURELLE write-back; architecture, CI/CD, Review 2 documents	WF-01, WF-04	D1, D3–D5; PR #3–#4, #11–#18, #20–#21
Lê Đức Trung Thi	Member	Returns and exchanges; stock receiving by lot (FIFO); 3D packing solver	WF-05, WF-06	D2; PR #7
Huỳnh Quốc Việt	Member	Packing Station UI (3D viewer, plan review, packing session); picking, shipping and returns screens	WF-03	PR #2, #8, #9, #10, #22
Phan Huỳnh Hải Phượng	Member	Admin portal (accounts, roles, leave); admin warehouse and SKU screens; marketplace connection screens	WF-02	Commits on feature/phuong_admin

Appendix A – Defense 1.1 Targets

A1. Main Workflows

ID	Workflow	Actors	Trigger	Business outcome	Features	Entity states	Owner
WF-01	Order Intake & Consolidation	Customer, Marketplace Platform (Lazada / AURELLE), OptiPack System, Store Owner / Shipping Coordinator, Warehouse Staff	Customer places an order on Lazada or AURELLE	Order imported, grouped by recipient, stock reserved, picker assigned	F-01, F-02, F-03	Order Group: awaiting_packaging → picking (or canceled)	Thuận
WF-02	Picking	Warehouse Staff, OptiPack System, Packaging Staff	Group in picking with an assigned picker	All items picked, or shortage decided (ship available / re-pick)	F-04, F-05	Order Group: picking → picked, or → partial_needs_review → picked / picking	Phượng
WF-03	Packing	OptiPack System, Packaging Staff, Reviewer (other staff / Admin)	Group reaches picked	Every parcel sealed and weighed; boxes and materials deducted	F-06, F-10	Packing Plan: computing → ready → approved → packing → packed (or rejected / failed / superseded); Order Group: picked → pending_approval → approved_for_packing → packed	Việt
WF-04	Shipping (GHN and Express Self-Delivery)	Shipping Coordinator, OptiPack System, GHN (Carrier), Customer, Warehouse Staff	Shipping Coordinator creates a shipment for a packed group	Parcels delivered, or returned to the warehouse with a failed-delivery ticket	F-07, F-10	Shipment: awaiting_pickup → in_transit → out_for_delivery → delivered / delivery_failed → returning_to_warehouse → returned_to_warehouse (or cancelled before pickup)	Thuận
WF-05	Return / Exchange via AURELLE	Customer, AURELLE Website, OptiPack System, Store Owner, Warehouse Staff	Customer submits a return / exchange request on AURELLE	Items inspected; refund recorded or replacement order created	F-08, F-09	Return Request: requested → awaiting_receipt → received → closed (or rejected / canceled)	Thi
WF-06	Stock Receiving by Lot	Warehouse Staff, OptiPack System, AURELLE Website	Goods arrive from the supplier	Stock received by lot; shortage groups re-checked; available stock pushed to AURELLE	F-04	Stock Lot created; stock-shortage flag of groups cleared	Thi

A2. Demo Scenarios

ID	Workflow	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-01	WF-01	Main	Lazada test shop connected; order for 2 SKUs in stock	Import → group → reserve → assign picker	Group in picking, picker assigned	Thuận
DS-02	WF-01	Exception	Order already cancelled when imported	Same flow with the cancelled order	Group cancelled automatically, reservation released; nothing is picked	Thuận
DS-03	WF-02	Main	Group in picking with stock in bins	Scan all items bin by bin → confirm	Group picked; oldest lots consumed	Phượng

ID	Workflow	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-04	WF-02	Exception	SKU not found in its bin	Report missing item	Group partial_needs_review; Store Owner and Packaging Staff notified	Phượng
DS-05	WF-03	Main	Picked group, SKU profiles confirmed	Compute plan → approve → scan → seal → weigh	Group packed	Việt
DS-06	WF-03	Exception	Parcel weight outside tolerance	Seal and weigh with a wrong weight	Parcel held and reviewed by another person	Việt
DS-07	WF-04	Main	Packed group; GHN staging configured	Quote → create shipment → GHN pickup → delivered	Group delivered; AURELLE customer sees every step	Thuận
DS-08	WF-04	Exception	GHN reports failed delivery and return (manual status buttons in the demo)	Fail → return → receive at warehouse	Shipment returned_to_warehouse; failed-delivery return ticket created	Thuận
DS-09	WF-05	Main	Delivered AURELLE order within 15 days	Request → approve → receive → inspect	Ticket closed; refund or replacement order	Thi
DS-10	WF-05	Exception	Request more than 15 days after delivery	Submit the request	Request refused with reason	Thi
DS-11	WF-06	Main	Warehouse with an empty bin	Assign SKU to bin → create receipt	Lot created; stock increased	Thi
DS-12	WF-06	Exception	Receipt exceeds bin capacity	Create the receipt	Receipt rejected with over-capacity detail	Thi

A3. Functional Test Case Target

Source	Count	Minimum per item	Subtotal
Mandatory features	10	3	30
Main workflows (end-to-end)	6	1	6
State transitions of main entities	33	1	33
Agreed NFRs	6	1	6
Committed total			75

Automated in CI: at least 50% of the committed total.